

Aaruush Data Vortex — Round 2

Evaluation Metrics Report — Sentiment Classification

Model Comparison

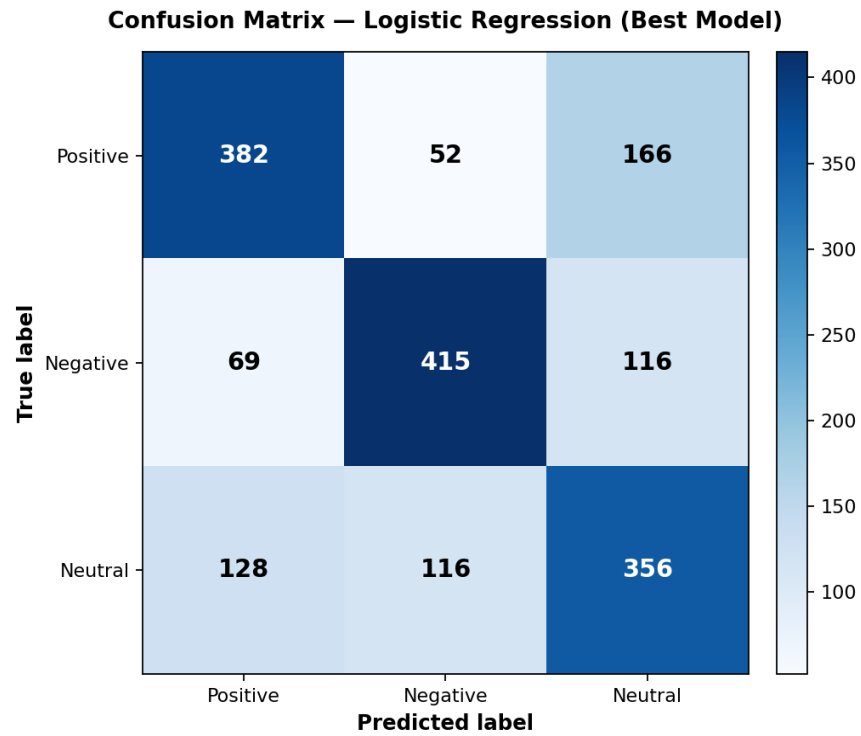
Model	Accuracy	Macro F1
Multinomial Naive Bayes	0.6128	0.6114
Logistic Regression (selected)	0.6406	0.6416
Linear SVM	0.6311	0.6319

Logistic Regression on TF-IDF features was selected: highest macro-F1 (0.642), which matters more than raw accuracy here since the 3 classes are perfectly balanced (3,000 each) but differ in difficulty, macro-F1 doesn't let strong performance on the easy classes mask weak performance on the hard one (Neutral).

Per-Class Metrics (Logistic Regression)

Class	Precision	Recall	F1-score	Support
Negative	0.712	0.692	0.702	600
Neutral	0.558	0.593	0.575	600
Positive	0.660	0.637	0.648	600
Macro avg	0.643	0.641	0.642	1800

Confusion Matrix



Overall accuracy: 64.1% on 1,800 held-out test posts (80/20 stratified split, 7,200 train / 1,800 test). Total misclassified: 647 (35.9%).